LETTERS
https://doi.org/10.1038/s41565-022-01154-9

nature nanotechnology

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Fischer–Tropsch synthesis to olefins boosted by MFI zeolite nanosheets

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Catalytic reactions are severely restricted by the strong adsorption of product molecules on the catalyst surface, where promoting desorption of the product and hindering its re-adsorption benefit the formation of free sites on the catalyst surface for continuous substrate conversion[1,2]. A solution to this issue is constructing a robust nanochannel for the rapid escape of products. We demonstrate here that MFI zeolite crystals with a short *b*-axis of 90–110 nm and a finely controllable microporous environment can effectively boost the Fischer–Tropsch synthesis to olefins by shipping the olefin molecules. The ferric carbide catalyst (Na-FeC$_x$) physically mixed with a zeolite promoter exhibited a CO conversion of 82.5% with an olefin selectivity of 72.0% at the low temperature of 260 °C. By contrast, Na-FeC$_x$ alone without the zeolite promoter is poorly active under equivalent conditions, and shows the significantly improved olefin productivity achieved through the zeolite promoter. These results show that the well-designed zeolite, as a promising promoter, significantly boosts Fischer–Tropsch synthesis to olefins by accelerating escape of the product from the catalyst surface.

Olefins have been used extensively as carbon-based building blocks for polymers and platform chemicals in the production of valuable oxygenates[3]. Current olefin production depends strongly on petroleum, and recently much attention has been focused on exploring alternative routes[4–8]. Among them, a promising blueprint has been to use the feedstock of syngas, a mixture of hydrogen and carbon monoxide (H_2 + CO) derived from biomass, coal and natural gas[6–15]. A classical syngas-to-olefin transformation is well known as the Fischer–Tropsch synthesis to olefins (FTO), which is normally performed at 320–400 °C using iron-based catalysts[15–22] because of their advantages of adaptability for the syngas with wide flexibility in H_2/CO ratios. A typical feature of the FTO is a wide product distribution, from C_1 up to even C_{20+} molecules following the Anderson–Schulz–Flory distribution, that usually contains abundant and undesirable methane (CH_4)[16,17]. In studying the FTO, much attention has been focused on innovating this process for the selective production of valuable olefins, such as ethene, propene or 1-hexene, at lower reaction temperatures although this has been a great challenge. For example, the Fe-based catalysts usually exhibit poor CO conversions at temperatures lower than 300 °C (ref. [6]), and higher temperatures are favourable for CO conversion[18] with a wide product distribution. In addition to the classical FTO route, the route based on an oxide–zeolite-based catalyst (OX-ZEO) was developed by combining a metal oxide and a zeolite (for example, ZnCrO$_x$ and SAPO-34) for a tandem conversion of syngas into methanol/ketene and further transformation to C_2–C_4 olefins[11,12]. Superior selectivity was obtained but the reactions were still performed at relatively high temperatures (~400 °C).

Here we report an efficient methodology to boost the activity and optimize the product selectivity for the low-temperature FTO (LT-FTO) over an Fe-based catalyst, which is achieved through mixing a finely designed zeolite as a promoter for olefin desorption from the catalyst surface (Fig. 1). In a proof-of-concept experiment, the sodium-doped ferric carbide catalyst (Na-FeC$_x$, with the Fe_5C_2 phase dominant, synthesized following the strategy reported previously; Supplementary Figs. 1 and 2) was used because it is one of the most classical catalysts for the FTO[6,9,23]. At 260 °C (syngas with a CO/H_2/Ar ratio of 45/45/10, a flow rate of 2,400 ml per h per g FeC$_x$, at 2.0 MPa pressure), this Na-FeC$_x$ catalyst showed a low activity with a CO conversion of 1.9% (Fig. 2a; Supplementary Fig. 3). An enhanced CO conversion of 25.6% was obtained by increasing the temperature to 300 °C, giving the C_1–C_{20+} products with a total olefin selectivity of 72.4% (Fig. 2a; Supplementary Table 1), which is similar to the results reported previously[16–18]. Interestingly, when the aluminosilicate MFI zeolite with a *b*-axis thickness of 90–110 nm (s-ZSM-5, with a Si/Al ratio of 26.2; Supplementary Fig. 4 and Supplementary Table 2) was mixed (a granule mixture with Na-FeC$_x$; Supplementary Fig. 5), the Na-FeC$_x$/s-ZSM-5 catalyst gave a CO conversion of 82.5% with an olefin selectivity of 72.0% at 260 °C (Fig. 2a,b); this gives an olefin productivity of 15.3 mmol per g FeC$_x$ per h, which is significantly higher than that of Na-FeC$_x$ under equivalent test conditions (Fig. 1). These data support the so-called LT-FTO, where the zeolite without any catalytic activity for CO transformation was shown to be a promoter, boosting the catalysis.

To explore the effect of the zeolite in LT-FTO, we studied the catalytic performance of Na-FeC$_x$ combined with different zeolites, which were shown to exhibit a higher CO conversion than that of bare Na-FeC$_x$ (Supplementary Fig. 3 and Supplementary Table 1). For example, the normal ZSM-5 zeolite (n-ZSM-5, with a Si/Al

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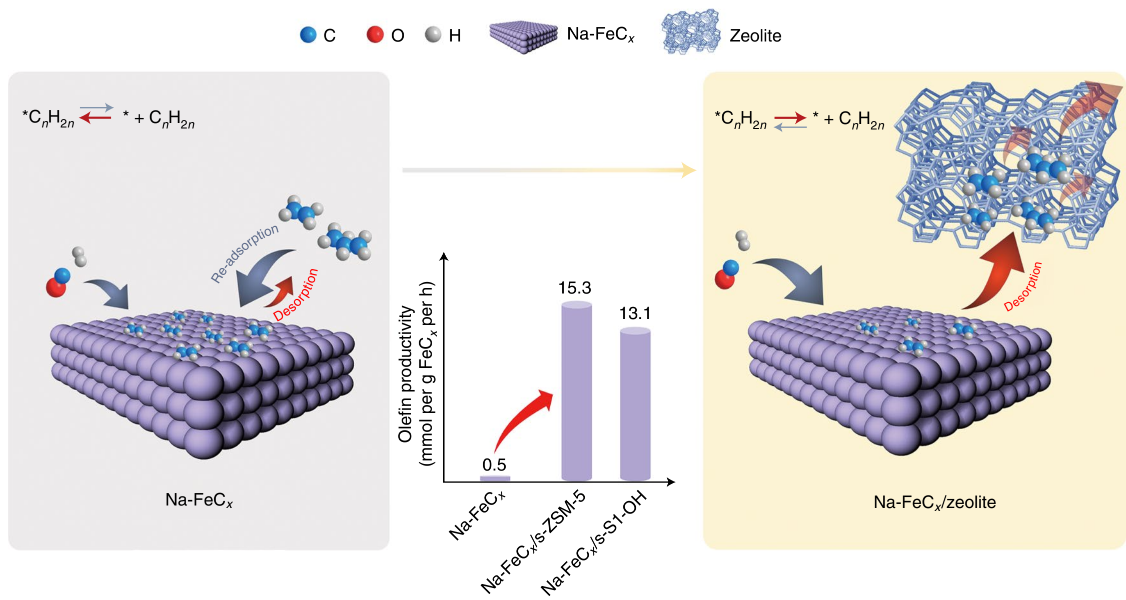

Fig. 1 | Schematic showing the strategy to boost the FTO via shifting the chemical equilibrium on the catalyst surface. In the model without the zeolite (left), the olefin products adsorbed on the Na-FeC$_x$ catalyst surface, as denoted by *, hinders CO hydrogenation to give a low olefin productivity. In the model with the zeolite (right), the olefin products rapidly escape from the zeolite micropores to form a free Na-FeC$_x$ surface for continuous CO hydrogenation. The olefin productivities over different catalysts are shown in the middle.

ratio of 30.5; Supplementary Fig. 6) combined with Na-FeC$_x$ (that is, Na-FeC$_x$/*n*-ZSM-5, with a zeolite/Na-FeC$_x$ mass ratio of 3) exhibited a CO conversion of 45.1% with C_2–C_{10} alkanes as the dominant products (an olefin/alkane molar ratio of only 0.1). The SSZ-13 zeolite (Supplementary Fig. 7) and the beta zeolite (Supplementary Fig. 8) mixed with the Na-FeC$_x$ improved the CO conversion to 39.8% and 32.5% with desired olefin/alkane (o/a) ratios of 4.2 and 4.1, respectively. The mordenite (or MOR) (Supplementary Fig. 9), SAPO-34 (Supplementary Fig. 10) and ZSM-22 zeolites (Supplementary Fig. 11), respectively, with Na-FeC$_x$ had a CO conversion of 62.2–74.1% and o/a ratios of 2.5–5.7. Among these zeolite-containing catalysts, the Na-FeC$_x$/s-ZSM-5 showed the highest CO conversion of 82.5% and an o/a ratio of 3.5 (Supplementary Figs. 12 and 13). By decreasing the zeolite/Na-FeC$_x$ ratio in the Na-FeC$_x$/s-ZSM-5 catalysts (with the zeolite/Na-FeC$_x$ mass ratio of 2), the o/a ratio could reach 4.2 with CO conversion at 66.8% (Supplementary Fig. 14 and Supplementary Table 3).

Regarding the product distribution, the Na-FeC$_x$-catalysed FTO reaction at 300 °C (that is, Na-FeC$_x$*) exhibited a wide carbon distribution of C_1–C_{20+} (Fig. 2b). By contrast, the LT-FTO over Na-FeC$_x$/s-ZSM-5 gave predominantly C_1–C_{10} molecules with limited C_{10+} products (Fig. 2b), indicating the significant advantage of LT-FTO for product selectivity compared with the conventional FTO. The LT-FTO narrows the product distribution to give predominantly C_2–C_{10} olefins (Supplementary Figs. 15 and 16), which include both the desired lower olefins (C_2–C_4) that are basic units for the modern chemical industry[9–12] and longer olefins that are valuable platform chemicals[18]. For example, 1-hexene and 1-octene are used as building units for high-performance polymers, and 1-pentene and 1-heptene can be converted into C_6/C_8 oxygenates via hydroformylation. In addition, this unique product distribution is also different from that of the OX-ZEO process, which predominantly produces lower (C_2–C_4) olefins[11,12,24–26].

The Na-FeC$_x$/s-ZSM-5 catalyst was evaluated using a continuous LT-FTO test. After activation for about 50 h, the CO conversion was constant in the steady state with an average value of about 68.1% (Fig. 2c). Such conversion was controlled to be lower than the aforementioned best result (82.5%) by adjusting the zeolite/Na-FeC$_x$ ratio, which is beneficial for evaluating the catalyst durability. Even after reaction for 600 h, the CO conversion was well maintained at 67.6%, with individual Na-FeC$_x$ and zeolite remaining stable (Supplementary Figs. 17 and 18). The selectivity for C_2–C_{10} olefins was also constant at 67.8–71.0% with an o/a ratio of 4.2–5.2. In these C_2–C_{10} olefin products, C_2–C_4 olefins and C_5–C_{10} olefins were about 45.3% and 54.7%, respectively (Supplementary Fig. 19). Notably, the selectivity for undesirable methane was lower than 3.8% (Fig. 2c). These data confirm the good durability of the Na-FeC$_x$/s-ZSM-5 catalyst in the LT-FTO reaction process. Normally, the carbon dioxide (CO_2) fraction is higher than 45% in the products, which is a general phenomenon in syngas-to-olefin reactions[6–8,15]. Owing to the combined advantages compared with classical FTO, which include being energy saving and using simpler equipment at a relatively low reaction temperature, as well as the high activity, optimized selectivity and good catalyst durability, we have compiled a process package based on this LT-FTO process to meet the demands for potential application (Supplementary Fig. 20).

Mixed catalysts that contain individual metal oxide and zeolite components have previously been studied in syngas conversion (for example, in OX-ZEO catalysts), where the zeolite catalyses the C–C coupling steps in cascade processes such as methanol-to-olefins reaction[11,12,24–27] and aromatization[28–30]. We emphasized that the zeolite in Na-FeC$_x$/s-ZSM-5 is conceptually different from the zeolite in the earlier composite catalysts, because the Na-FeC$_x$ can directly catalyse the syngas-to-olefin reaction[6,18]. Although the Na-FeC$_x$/s-ZSM-5 catalyst indeed controlled the product selectivity so that C_1–C_{10} products were dominant compared with the C_1–C_{20+} products that dominate on bare Na-FeC$_x$, this feature should not be assigned to acid-catalysed reactions, such as the cracking of large olefins formed on the Na-FeC$_x$. This possibility can be excluded in explaining the unique selectivity of LT-FTO, because the cracking of large olefins would produce isomerized olefins that cannot meet the high α-olefin selectivity in the Na-FeC$_x$/s-ZSM-5-catalysed LT-FTO, as confirmed by the model cracking reactions of 1-butene, 1-octene and 1-hexene over s-ZSM-5 at 260 °C (Supplementary Fig. 21a and Supplementary Tables 4–6). Therefore, the final α-olefin products in LT-FTO would result from the FTO step on Na-FeC$_x$ rather

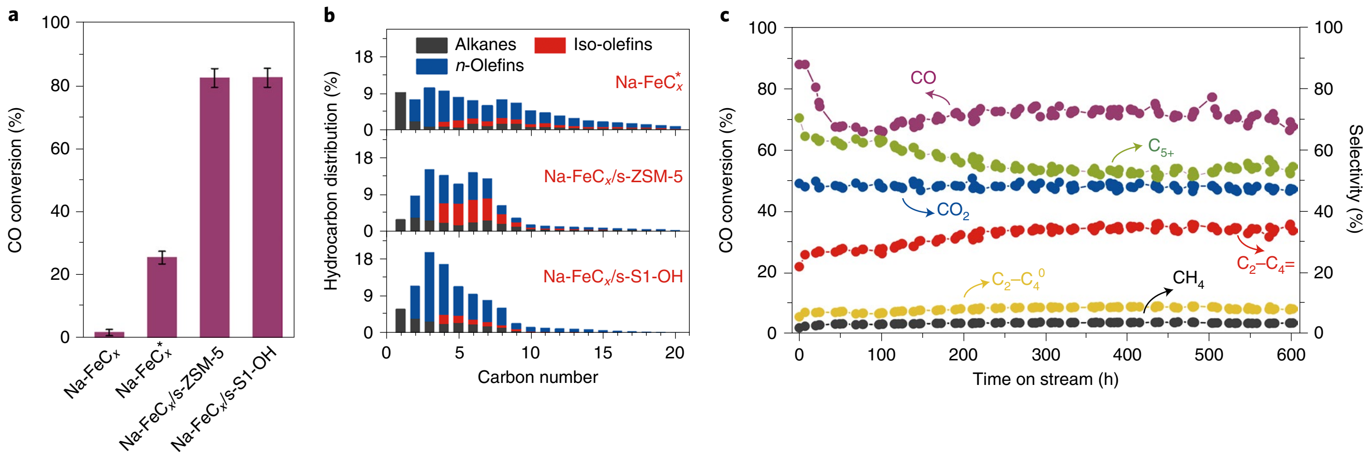

Fig. 2 | Catalytic performance of the Na-FeC$_x$/zeolite catalysts in the LT-FTO. a,**b**, CO conversion (**a**) and hydrocarbon distribution (**b**) for Na-FeC$_x$, Na-FeC$_x$/s-ZSM-5 and Na-FeC$_x$/s-S1-OH in FTO. Reaction conditions: zeolite:Na-FeC$_x$ weight ratio, 3; H_2/CO ratio, 1; gas flow rate, 2,400 ml per h per g FeC$_x$; reactor pressure, 2.0 MPa, temperature, 260 °C; granule mixing of the individual zeolite and Na-FeC$_x$ components. Na-FeC$_x$ marked with * denotes the reaction at 300 °C. For Na-FeC$_x$/s-S1-OH the H_2/CO molar ratio in the feed gas was 3. CO_2 was not included in the product distribution. **c**, Catalytic data for the Na-FeC$_x$/s-ZSM-5 catalyst in a continuous LT-FTO test. Reaction conditions: s-ZSM-5:Na-FeC$_x$, 2; H_2/CO, 1; 2,400 ml per h per g FeC$_x$; 2.0 MPa; 260 °C; granule mixing of the individual zeolite and Na-FeC$_x$ components. CO_2 was not included in calculating the selectivity for hydrocarbon products. C_2–C_4^0, lower alkanes; C_2–$C_4^=$, lower olefins; C_{5+}, C_5 products or higher.

than from acid-catalysed steps. For the production of α-olefins, s-ZSM-5 can serve as a promoter rather than a tandem acid catalyst. This feature is different from the role of zeolites in previous metal oxide/carbide and zeolite composite catalysts, where those zeolites act as acid catalysts for aromatization, isomerization, cracking and C–C coupling reactions, such as in the OX-ZEO system, where the final products are all produced via acid-catalysed reactions (Supplementary Fig. 21b)[11,12,24–26,31–37].

To confirm this concept and completely avoid the influence of acid sites on the aluminosilicate zeolite (for example, s-ZSM-5), we rationally designed a siliceous zeolite with a controlled *b*-axis thickness of 90–110 nm and artificially introduced silanol groups on the framework using an organosilane-assisted method[38,39], which is denoted as s-S1-OH (Supplementary Figs. 22–26). The Na-FeC$_x$/s-S1-OH catalyst showed a remarkably enhanced CO conversion of 66.5% compared with 1.9% on the bare Na-FeC$_x$ under equivalent test conditions (syngas with a CO/H_2/Ar ratio of 45/45/10, 2,400 ml per h per g FeC$_x$, 2.0 MPa; Supplementary Figs. 27 and 28 and Supplementary Table 7), exhibiting a olefin productivity of 13.1 mmol per g FeC$_x$ per h. By adjusting the H_2/CO ratio to 3, the CO conversion could even reach 82.6% in the LT-FTO (Fig. 2a; Supplementary Fig. 29 and Supplementary Table 8). Importantly, Na-FeC$_x$/s-S1-OH still showed a narrow distribution that was dominated by C_1–C_{10} molecules with an o/a ratio of 3.8, where the methane and C_{10+} product selectivities are 6.0% and 6.3% (Supplementary Table 8), respectively. Considering that the siliceous zeolite has no active sites for reactions in the LT-FTO process (Supplementary Fig. 30), the promoter role of the zeolite is confirmed in the Na-FeC$_x$-catalysed reactions. Moreover, the Na-FeC$_x$/s-S1-OH catalyst showed that 81.7% of the C_{4+} olefin products are α-olefins, which is much higher than the 52.1% seen over the Na-FeC$_x$/s-ZSM-5 catalyst (Fig. 2b), which is due to the s-S1-OH zeolite having no Brønsted acid sites that might cause undesirable α-olefin isomerization (Supplementary Fig. 21a). Although ferric-based catalysts for the FTO have been investigated for many years, it remains a challenge to simultaneously achieve both a high CO conversion and a high olefin selectivity at a given temperatures (for example, 260 °C). These data confirm the good performance of the Na-FeC$_x$/zeolite catalyst[10].

The influence of the mixing method for Na-FeC$_x$ and the zeolite in the LT-FTO process was also studied (Fig. 3a; Supplementary Fig. 31 and Supplementary Tables 9 and 10). Compared with granule mixing in the aforementioned cases, a powder mixture—prepared by mixing Na-FeC$_x$ and s-ZSM-5 powdered samples together and then made into granules for catalytic tests—exhibited a much lower CO conversion (21.5%) with a large amount of alkane by-products (o/a ratio 1.3). The selectivity for undesirable methane reached 18.8%, much higher than the 3.0% seen over the Na-FeC$_x$/s-ZSM-5 granule-mixing catalyst. This phenomenon could be due to massive sodium migration from Na-FeC$_x$ to the zeolite in the powder-mixing catalyst, as confirmed via Na 1*s* X-ray photoelectron spectroscopy characterization (Supplementary Figs. 32 and 33), which resulted in Na-FeC$_x$ having an insufficient sodium doping level to give a performance similar to the classical Fischer–Tropsch synthesis to alkanes. In addition, the nanometre-scale proximity in the powder-mixing catalyst might enhance hydrogen spillover in the catalyst bed that also benefits deep hydrogenation[39–41]. Using the dual-bed approach, s-ZSM-5 was packed below the Na-FeC$_x$ bed, separated by a layer of inert quartz sand, resulting in a poor CO conversion. In Na-FeC$_x$/s-ZSM-5-catalysed LT-FTO, raising the gas flow rate from 2,400 to 9,600 ml per h per g FeC$_x$ efficiently decreased the CO conversion but showed a similar product selectivity (Supplementary Fig. 34 and Supplementary Table 11). A similar phenomenon was also observed using the Na-FeC$_x$/s-S1-OH catalyst (Supplementary Fig. 35 and Supplementary Table 12).

To understand the promotion effect of the zeolite, we explored the essential factors for controlling the FTO reaction. As shown in Fig. 3b, the Na-FeC$_x$ catalyst gave a CO conversion of 29.9% in the FTO at 300 °C. When ethene was co-introduced with the feed gas (at 16.7 vol%), about 88% of the activity was lost, showing a CO conversion of 3.5% after the introduction of ethene for 5 h. Supplementary Figure 36 shows the dependence of the CO conversion on the ethene concentration for this FTO reaction, and gives lower activity with more ethene in the feed gas. This result can be explained by the known feature of the ferric carbide catalyst, whose surface strongly adsorbs the olefin molecules, causing difficult-to-desorb species to block the active sites[23,42,43]. When ethene was removed from the feed gas, the CO conversion could be regenerated to 28.8% (Fig. 3b).

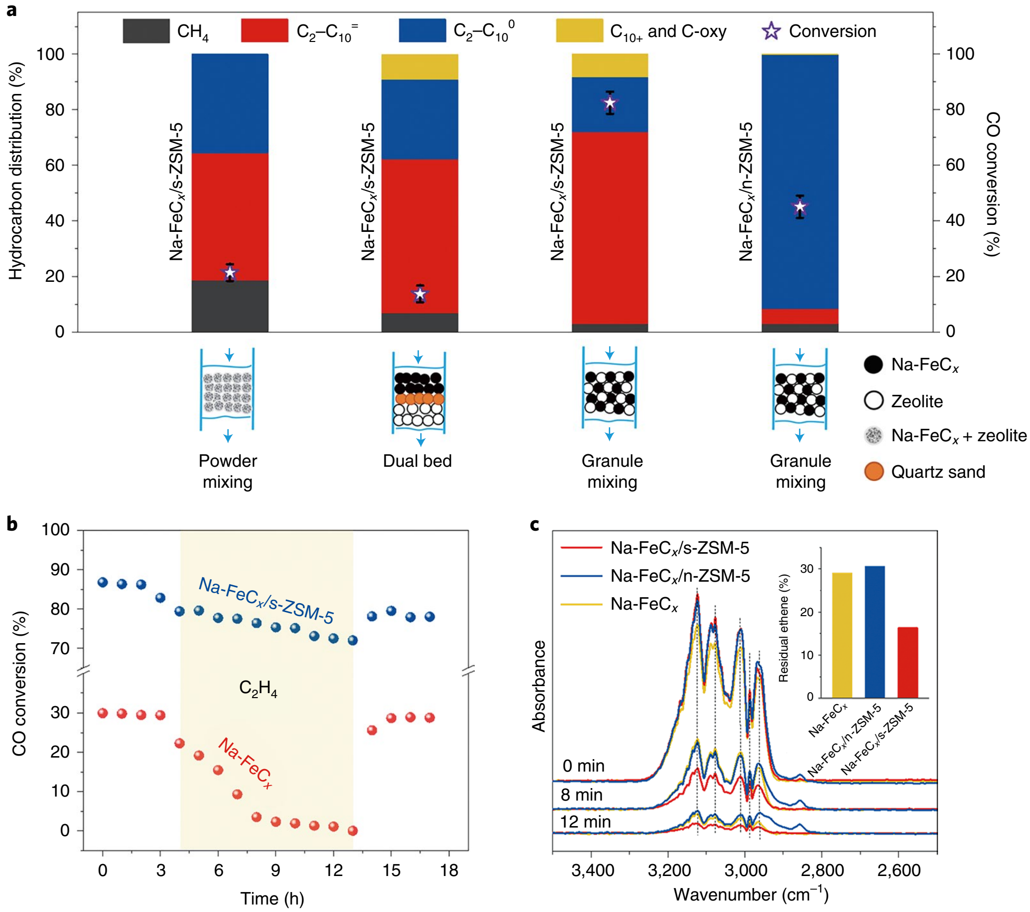

Fig. 3 | Catalytic data for the Na-FeC$_x$/zeolite catalysts in LT-FTO and ethene desorption DRIFT spectra over Na-FeC$_x$/zeolite catalysts. a, Data showing the performance of catalysts, using different mixing methods, in the LT-FTO. Reaction conditions: zeolite:Na-FeC$_x$ weight ratio, 3; H_2/CO ratio, 1; flow rate, 2,400 ml per h per g FeC$_x$; pressure, 2.0 MPa; temperature, 260 °C. CO_2 was not included in the product distribution. C_2–C_{10}^0, C_2–C_{10} alkanes; C_2–$C_{10}^=$, C_2–C_{10} olefins; C_{10+}, C_{10} products or higher; C-oxy, oxygenated carbon compounds. Error bounds for CO conversions were ±2%. **b**, Data showing the influence of ethene (C_2H_4) on the LT-FTO using the Na-FeC$_x$/s-ZSM-5 and Na-FeC$_x$ catalysts. Reaction conditions: s-ZSM-5:Na-FeC$_x$, 3; ethene/H_2/CO, 0.6/2.0/1.0; 2,880 ml per h per g FeC$_x$; 2.0 MPa, granule mixing of the individual s-ZSM-5 and Na-FeC$_x$ components for Na-FeC$_x$/s-ZSM-5; reaction temperature, 260 °C for Na-FeC$_x$/s-ZSM-5 and 300 °C for Na-FeC$_x$. **c**, Ethene desorption DRIFT spectra over the Na-FeC$_x$/s-ZSM-5, Na-FeC$_x$/n-ZSM-5 and Na-FeC$_x$ catalysts. The inset shows the residual percentage of ethene according to the peak-intensity changes (signal at ~2,986 cm^{-1}) over the different catalysts for 0–12 min (data calculated from Supplementary Table 13).

These results help us to understand that the poor activity of the Na-FeC$_x$ catalyst in the LT-FTO might be due to the difficulty of olefin product desorption at lower temperatures (Supplementary Fig. 37). By contrast, the equivalent introduction of ethene causes only a slight loss of activity over the Na-FeC$_x$/s-ZSM-5 and Na-FeC$_x$/s-S1-OH catalysts, at even lower temperature, confirming the improved olefin tolerance of Na-FeC$_x$ when mixed with s-ZSM-5 or s-S1-OH zeolites (Fig. 3b; Supplementary Fig. 38). It is possible that these rationally designed zeolites help desorption of the olefin products once they have been formed on the Na-FeC$_x$ surface (Supplementary Fig. 37), thus improving the activity.

This hypothesis was supported by diffuse reflectance infrared Fourier transform (DRIFT) spectra characterizing the olefin desorption on various catalysts including Na-FeC$_x$, Na-FeC$_x$/n-ZSM-5 and Na-FeC$_x$/s-ZSM-5. As shown in Fig. 3c, after pre-adsorption treatment of ethene as a model, all samples showed the bands which were assigned to the symmetrical and asymmetrical stretching vibration of $=CH_2$ (ν_s=CH_2 signals at 2,966, 2,986, 3,011 and 3,124 cm^{-1}; ν_{as}=CH_2 signal at 3,075 cm^{-1})[44–46]. Introducing an argon (Ar) flow to the samples, the signals were reduced because of the ethene desorption. According to the peak-intensity changes as a function of time, the ethene desorption rates followed the order of Na-FeC$_x$/s-ZSM-5 > Na-FeC$_x$ > Na-FeC$_x$/n-ZSM-5 (Supplementary Figs. 39–42 and Supplementary Table 13), which is confirmed in the equivalent infrared spectroscopy tests using propene as a probing molecule (Supplementary Figs. 43–47 and Supplementary Table 14). These data demonstrate that the s-ZSM-5 zeolite indeed accelerates the desorption of olefins once they are formed on the Na-FeC$_x$ surface via C–C coupling in the FTO, which is favourable for the continuous hydrogenation of CO. In addition, the rapid olefin desorption also inhibits the growth of carbon chains to optimize the selectivity, hindering the formation of long-chain hydrocarbons. The s-S1-OH zeolite displayed the same behaviour as s-ZSM-5 in accelerating the olefin desorption, as shown in the DRIFT spectra of the samples (Supplementary Figs. 48 and 49). By contrast, the n-ZSM-5 zeolite delayed the olefin desorption on the Na-FeC$_x$/n-ZSM-5 catalyst. The long retention time of olefins in the Na-FeC$_x$/n-ZSM-5 catalyst led to olefins being hydrogenated (Supplementary Fig. 50), which could explain the large amount of alkane products rather than olefins in the LT-FTO over the Na-FeC$_x$/n-ZSM-5 catalyst (Fig. 3a).

In previous studies using zeolite and sodium-modified oxide/carbide mixtures, sodium migration may occur to influence the

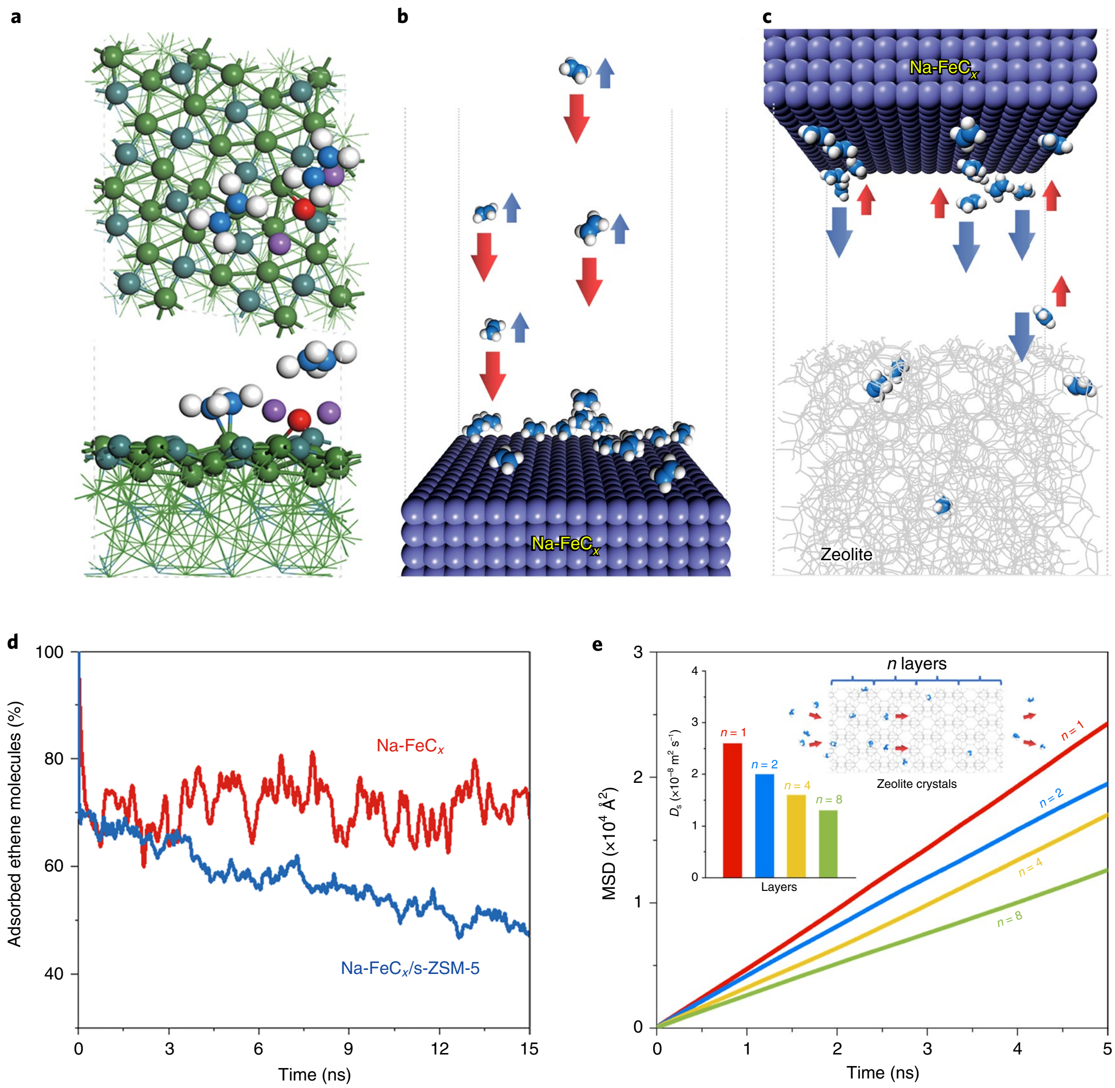

Fig. 4 | Computational models, DFT calculations, MD simulations and the MSD of ethene. a, Top and side view of the adsorption model showing ethene molecules on the Na-FeC$_x$ surface (C_2H_4: C atoms in blue and H atoms in white; Na-FeC$_x$: O atoms in red, Na atoms in purple, Fe atoms in green and C atoms in dark cyan). **b**,**c**, Schematics showing the adsorption–desorption equilibrium of ethene molecules on Na-FeC$_x$ (**b**) and Na-FeC$_x$/s-ZSM-5 (**c**). **d**, Percentage of ethene molecules adsorbed on the Na-FeC$_x$ surface during the desorption process over Na-FeC$_x$ and Na-FeC$_x$/s-ZSM-5. **e**, MSD of ethene molecules diffusing in s-ZSM-5 zeolites with different numbers of unit layers (n). Insets: the model (right) and D_s values (left) in the MFI zeolite micropores with one, two, four and eight unit layers.

intrinsic catalytic activity of the individual oxide/carbide compounds[31,33,36,47,48]. Notably, this hypothesis can be excluded to explain the high CO conversion on the Na-FeC$_x$/zeolite catalysts in the LT-FTO (Supplementary Figs. 51 and 52 and Supplementary Tables 10 and 15). Na-FeC$_x$ recycled from the used Na-FeC$_x$/s-ZSM-5 catalyst still exhibited a comparable performance to that of fresh Na-FeC$_x$ (Supplementary Table 15). Therefore, the fast desorption of olefins from the catalyst surface should dominate the accelerated reaction. Olefin desorption from the composite catalysts should proceed via efficient olefin transfer from the Na-FeC$_x$ surface to the zeolite micropores and then rapidly escape from the zeolite micropores. To understand this process, we performed theoretical simulations to provide a deeper insight into the adsorption and diffusion characteristics of ethene on the Na-FeC$_x$ surface as a model (Fig. 4a; Supplementary Fig. 53). In practice, the free olefins may either escape to form the final product or re-adsorb on the Na-FeC$_x$ surface to participate an adsorption–desorption equilibrium ($^*C_2H_4 \leftrightarrows {}^* + C_2H_4$), which hinders the escape of ethene (Fig. 4b; Supplementary Fig. 54). Figure 4c shows the model containing the Na-FeC$_x$ surface and adjacent zeolite framework with a free region between them (Supplementary Fig. 55), for considering the separated Na-FeC$_x$ and s-ZSM-5 components in the practical catalytic tests. Using this model, the ethene molecules on the Na-FeC$_x$ surface desorb into the free region, where these free olefins can be rapidly adsorbed by the zeolite to shift the sorption equilibrium on Na-FeC$_x$ and accelerate the desorption of more ethene molecules (Supplementary Fig. 56). The ethene desorption processes were qualitatively predicted on the Na-FeC$_x$ surface as a function of the desorption time using molecular dynamics (MD) simulations at 260 °C (Fig. 4d). It was observed that the number of ethene molecules on the Na-FeC$_x$ surface maintained a quantitative equilibrium of about 71% for the model without the zeolite. By contrast, this number clearly decreased to about 48% for the model with the zeolite, confirming the shifted equilibrium between desorption and re-adsorption in the presence of the microporous ZSM-5 zeolite. In this case, the zeolite did not directly interact with the olefin molecules on the Na-FeC_x surface, but captured the olefins in the free region to shift the equilibrium (Supplementary Figs. 57 and 58).

With regard to ethene molecules in the zeolite crystals, density functional theory (DFT) calculations showed adsorption energies of −0.42, −0.78 and −0.59 eV within the micropores for S-1, ZSM-5 and S1-OH, respectively (Supplementary Fig. 59), which confirmed the enhanced adsorption of ethene by the acid sites or silanol groups within the zeolite. This result has been illustrated by the well-known H–π interaction between a proton and a C=C bond[49], confirming the importance of the zeolite microporous environment for olefin adsorption. We also investigated the ethene molecule diffusing along the straight channels of the MFI zeolite structure with different unit layers (Supplementary Fig. 60). Interaction energy profiles showed that the diffusion barriers for ethene are strongly related to the adsorption sites, and the diffusion efficiency in one zeolite crystal should be inversely proportional to the number of adsorption sites and the zeolite channel distance, which can be adjusted via the MFI zeolite crystal morphology (for example, s-ZSM-5 versus n-ZSM-5 have a similar composition but a different morphology). Further studies on olefin diffusion in the zeolite micropores were performed, and from the slope of the mean square displacement (MSD) values of the ethene diffusion coefficient (D_s) with various zeolite layers were quantitatively determined (Fig. 4e; Supplementary Fig. 61)[50]. The D_s values were 2.6×10^{-8}, 2.0×10^{-8}, 1.6×10^{-8} and $1.3\times10^{-8}\,m^2\,s^{-1}$ in the s-ZSM-5 MFI zeolite with one, two, four and eight unit layers, respectively. These results suggest that thinner zeolite crystals with shorter diffusion distances have a significant advantage for the more rapid shipping of olefin products, resulting in short retention times of the olefin molecules in the zeolite crystals; this will be beneficial for the fast formation of free micropores for continuously adsorbing more olefin products. To further address this issue, we theoretically simulated the olefin product desorption from the Na-FeC_x surface using a zeolite with residual olefins in the micropores. In this case, the number of ethene molecules on the Na-FeC_x surface maintained a quantitative equilibrium of about 69%, which is much higher than 48% for the case of using zeolite with free micropores (Supplementary Fig. 62). This outcome is reasonable due to the lack of sufficient free micropores on the MFI zeolite because they are blocked by residual olefin molecules, confirming the importance of rapid olefin shipping in zeolite crystals.

We have demonstrated the LT-FTO process with boosted activity and optimized selectivity on a classical ferric carbide catalyst combined with a zeolite promoter via appropriate mixing. Multiple experimental and theoretical studies revealed that the zeolite, with a reasonably controlled microporous environment for adsorbing olefins and a short *b*-axis thickness for the shipping of olefins, can accelerate the escape of product molecules. This feature benefits the formation of a free catalyst surface for CO hydrogenation. We believe that the designed zeolites can act as a powerful promoter for syngas conversion. In addition to the equilibrium change, other interpretations of the zeolite promotion effect should be considered in future work for deeply understanding the synergism between metal carbide and zeolite, which could open an alternative route to the design and preparation of more efficient heterogeneous catalysts for syngas conversion.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41565-022-01154-9.

Received: 11 September 2021; Accepted: 23 May 2022;
Published online: 11 July 2022

References

1. Li, H. et al. Na^+-gated water-conducting nanochannels for boosting CO_2 conversion to liquid fuels. *Science* **367**, 667–671 (2020).
2. Morejudo, S. H. et al. Direct conversion of methane to aromatics in a catalytic co-ionic membrane reactor. *Science* **353**, 563–566 (2016).
3. Ren, T., Patel, M. & Rlok, K. Olefins from conventional and heavy feedstocks: energy use in steam cracking and alternative processes. *Energy* **31**, 425–451 (2006).
4. Snel, R. Olefins from syngas. *Catal. Rev. Sci. Eng.* **29**, 361–445 (1987).
5. Dry, M. E. The Fischer–Tropsch process: 1950–2000. *Catal. Today* **71**, 227–241 (2002).
6. Torres Galvis, H. M. & de Jong, K. P. Catalysts for production of lower olefins from synthesis gas: a review. *ACS Catal.* **3**, 2130–2149 (2013).
7. Pan, X., Jiao, F., Miao, D. & Bao, X. Oxide–zeolite-based composite catalyst concept that enables syngas chemistry beyond Fischer–Tropsch synthesis. *Chem. Rev.* **121**, 6588–6609 (2021).
8. Zhou, W. et al. New horizon in C1 chemistry: breaking the selectivity limitation in transformation of syngas and hydrogenation of CO into hydrocarbon chemicals and fuels. *Chem. Soc. Rev.* **48**, 3193–3228 (2019).
9. Galvis, H. M. T. et al. Supported iron nanoparticles as catalysts for sustainable production of lower olefins. *Science* **335**, 835–838 (2012).
10. Zhong, L. et al. Cobalt carbide nanoprisms for direct production of lower olefins from syngas. *Nature* **538**, 84–87 (2016).
11. Jiao, F. et al. Selective conversion of syngas to light olefins. *Science* **351**, 1065–1068 (2016).
12. Cheng, K. et al. Direct and highly selective conversion of synthesis gas into lower olefins: design of a bifunctional catalyst combining methanol synthesis and carbon–carbon coupling. *Angew. Chem. Int. Ed.* **55**, 4725–4728 (2016).
13. Wang, P. et al. Synthesis of stable and low-CO_2 selective ε-iron carbide Fischer–Tropsch catalysts. *Sci. Adv.* **4**, eaau2947 (2018).
14. Li, J. et al. Integrated tuneable synthesis of liquid fuels via Fischer–Tropsch technology. *Nat. Catal.* **1**, 787–793 (2018).
15. Xu, Y. et al. A hydrophobic FeMn@Si catalyst increases olefins from syngas by suppressing C1 by-products. *Science* **371**, 610–613 (2021).
16. Soled, S., Iglesia, E. & Fiato, R. A. Activity and selectivity control in iron catalyzed Fischer–Tropsch synthesis. *Catal. Lett.* **7**, 271–280 (1990).
17. Shroff, M. D. et al. Activation of precipitated iron Fischer–Tropsch synthesis catalysts. *J. Catal.* **156**, 185–207 (1995).
18. Zhai, P. et al. Highly tunable selectivity for syngas-derived alkenes over zinc and sodium-modulated Fe_5C_2 catalyst. *Angew. Chem. Int. Ed.* **55**, 9902–9907 (2016).
19. Koeken, A. C. J., Torres Galvis, H. M., Davidian, T., Ruitenbeek, M. & de Jong, K. P. Suppression of carbon deposition in the iron-catalyzed production of lower olefins from synthesis gas. *Angew. Chem. Int. Ed.* **51**, 7190–7193 (2012).
20. Torres Galvis, H. M. et al. Iron particle size effects for direct production of lower olefins from synthesis gas. *J. Am. Chem. Soc.* **134**, 16207–16215 (2012).
21. Liu, Y., Chen, J. F., Bao, J. & Zhang, Y. Manganese-modified Fe_3O_4 microsphere catalyst with effective active phase of forming light olefins from syngas. *ACS Catal.* **5**, 3905–3909 (2015).
22. Lohitharn, N., Goodwin, J. G. Jr. & Lotero, E. Fe-based Fischer–Tropsch synthesis catalysts containing carbide-forming transition metal promoters. *J. Catal.* **255**, 104–113 (2008).
23. de Smit, E. & Weckhuysen, B. M. The renaissance of iron-based Fischer–Tropsch synthesis: on the multifaceted catalyst deactivation behaviour. *Chem. Soc. Rev.* **37**, 2758–2781 (2008).
24. Jiao, F. et al. Shape-selective zeolites promote ethylene formation from syngas via a ketene intermediate. *Angew. Chem. Int. Ed.* **57**, 4692–4696 (2018).
25. Zhu, Y. et al. Role of manganese oxide in syngas conversion to light olefins. *ACS Catal.* **7**, 2800–2804 (2017).
26. Liu, X. et al. Tandem catalysis for hydrogenation of CO and CO_2 to lower olefins with bifunctional catalysts composed of spinel oxide and SAPO-34. *ACS Catal.* **10**, 8303–8314 (2020).
27. Zhu, X. et al. Trimodal porous hierarchical SSZ-13 zeolite with improved catalytic performance in the methanol-to-olefins reaction. *ACS Catal.* **6**, 2163–2177 (2016).
28. Zhao, B. et al. Direct transformation of syngas to aromatics over Na-Zn-Fe_5C_2 and hierarchical HZSM-5 tandem catalysts. *Chem* **3**, 323–333 (2017).
29. Cheng, K. et al. Bifunctional catalysts for one-step conversion of syngas into aromatics with excellent selectivity and stability. *Chem* **3**, 334–347 (2017).
30. Yang, J., Pan, X., Jiao, F., Li, J. & Bao, X. Direct conversion of syngas to aromatics. *Chem. Commun.* **53**, 11146–11149 (2017).
31. Botes, F. G. & Böhringer The addition of HZSM-5 to the Fischer–Tropsch process for improved gasoline production. *Appl. Catal. A Gen.* **267**, 217–225 (2004).
32. Gwagwa, X. Y. & van Steen, E. Migration of potassium in an Fe_2O_3/H-ZSM-5 composite catalyst. *Chem. Eng. Technol.* **32**, 826–829 (2009).
33. Karre, A. V., Kababji, A., Kugler, E. L. & Dadyburjor, D. B. Effect of addition of zeolite to iron-based activated-carbon-supported catalyst for Fischer–Tropsch synthesis in separate beds and mixed beds. *Catal. Today* **198**, 280–288 (2012).

34. Karre, A. V., Kababji, A., Kugler, E. L. & Dadyburjor, D. B. Effect of time on stream and temperature on upgraded products from Fischer–Tropsch synthesis when zeolite is added to iron-based activated-carbon-supported catalyst. *Catal. Today* **214**, 82–89 (2013).
35. Li, B. et al. In-situ crystallization route to nanorod-aggregated functional ZSM-5 microspheres. *J. Am. Chem. Soc.* **135**, 1181–1184 (2013).
36. Weber, J. L. et al. Effect of proximity and support material on deactivation of bifunctional catalysts for the conversion of synthesis gas to olefins and aromatics. *Catal. Today* **342**, 161–166 (2020).
37. Weber, J. L. et al. Conversion of synthesis gas to aromatics at medium temperature with a Fischer Tropsch and ZSM-5 dual catalyst bed. *Catal. Today* **369**, 175–183 (2021).
38. Wang, C. et al. Importance of zeolite wettability for selective hydrogenation of furfural over Pd@Zeolite catalysts. *ACS Catal.* **8**, 474–481 (2018).
39. Wang, C. et al. Product selectivity controlled by nanoporous environments in zeolite crystals enveloping rhodium nanoparticle catalysts for CO_2 hydrogenation. *J. Am. Chem. Soc.* **141**, 8482–8488 (2019).
40. Im, J., Shin, H., Jang, H., Kim, H. & Choi, M. Maximizing the catalytic function of hydrogen spillover in platinum-encapsulated aluminosilicates with controlled nanostructures. *Nat. Commun.* **5**, 3370 (2014).
41. Wang, S. et al. Activationand spillover of hydrogen on sub-1 nm palladium nanoclusters confined within sodalite zeolite for the semi-hydrogenation of alkynes. *Angew. Chem. Int. Ed.* **58**, 7668–7672 (2019).
42. Niemantsverdriet, J. W., der Kraan, A. M. V., Dijk, W. L. W. & der Baan, H. S. V. Behavior of metallic iron catalysts during Fischer–Tropsch synthesis studied with Mössbauer spectroscopy, X-ray diffraction, carbon content determination, and reaction kinetic measurements. *J. Phys. Chem.* **84**, 3363–3370 (1980).
43. Li, S., Li, A., Krishnamoorthy, S. & Iglesia, E. Effects of Zn, Cu, and K promoters on the structure and on the reduction, carburization, and catalytic behavior of iron based Fischer–Tropsch synthesis catalysts. *Catal. Lett.* **77**, 197–205 (2001).
44. Efremov, A. A. & Davydov, A. A. Infrared spectra of π-complexes of propylene and ethylene on TiO_2. *React. Kinet. Catal. Lett.* **15**, 327–331 (1980).
45. Ji, W., Chen, Y., Shen, S., Li, S. & Wang, H. FTIR study of adsorption of CO, NO and C_2H_4 and reaction of CO + H_2 on the well-dispersed FeO_xγ-Al_2O_3 and FeO_x/TiO_2(a) catalysts. *Appl. Surf. Sci.* **99**, 151–160 (1996).
46. Leclerc, H. et al. Infrared study of the influence of reducible iron(III) metal sites on the adsorption of CO, CO_2, propane, propene and propyne in the mesoporous metal–organic framework MIL-100. *Phys. Chem. Chem. Phys.* **13**, 11748–11756 (2011).
47. Li, M., Nawaz, M. A., Song, G., Zaman, W. Q. & Liu, D. Influential role of elemental migration in a composite iron–zeolite catalyst for the synthesis of aromatics from syngas. *Ind. Eng. Chem. Res.* **59**, 9043–9054 (2020).
48. Wang, T. et al. Sodium-mediated bimetallic Fe–Ni catalyst boosts stable and selective production of light aromatics over HZSM-5 zeolite. *ACS Catal.* **11**, 3553–3574 (2021).
49. Cnudde, P. et al. Experimental and theoretical evidence for the promotional effect of acid sites on the diffusion of alkenes through small-pore zeolites. *Angew. Chem. Int. Ed.* **60**, 10016–10022 (2021).
50. Smit, B. & Maesen, T. L. M. Molecular simulations of zeolites: adsorption, diffusion, and shape selectivity. *Chem. Rev.* **108**, 4125–4184 (2008).

Methods

Synthesis of s-ZSM-5 zeolite. The ZSM-5 zeolite with a sheet morphology and a *b*-axis thickness of 90–110 nm (denoted as s-ZSM-5, where 's' stands for sheet) was synthesized using a urea-assisted method. As a typical run, tetrapropylammonium hydroxide (TPAOH; 4.0 g, 25% aqueous solution) was dissolved in deionized water (5.1 g) with stirring for 0.5 h, and then tetraethylorthosilicate (TEOS; 2.8 g) was added to the solution with stirring. After the addition of aluminium isopropoxide (0.07 g) and urea (0.65 g), the mixture was stirred continuously at room temperature for another 12 h. The resulting gel, with the molar composition of SiO_2/Al_2O_3/TPAOH/H_2O/urea of 1/0.0128/0.3684/33.8346/0.812, respectively, was transferred into an autoclave for crystallization at 180 °C for two days. The as-synthesized products were collected by filtration, washed with water, dried in air and calcined at 550 °C for 4 h to remove the organic template. Finally, the s-ZSM-5 zeolite with a Si/Al ratio of 26.2 was obtained. For this work, unless specified otherwise, the s-ZSM-5 zeolite is in the protonated form. For comparison, s-ZSM-5 zeolites with Si/Al ratios of 16.5, 84.3 and 141.2 were synthesized via similar procedures using 0.14, 0.035 and 0.02 g of aluminium isopropoxide in the starting gels, respectively. In this work, unless specified otherwise, the s-ZSM-5 zeolite has the Si/Al ratio of 26.2.

Synthesis of s-S1-OH zeolite. The S-1 zeolite with a sheet morphology, artificially introduced silanol groups and *a b*-axis thickness of 90–115 nm (denoted as s-S1-OH, where s stands for sheet) was synthesized using a urea-assisted method in the presence of organosilane in the starting gel. As a typical run, TPAOH (4.0 g, 25% aqueous solution) was dissolved in deionized water (5.1 g) with stirring for 0.5 h, and then TEOS (2.52 g) and diethoxydimethylsilane (DEMS; 0.199 g) was added to the solution with stirring (where the molar ratio of organosilane to the total amount of silica was 1:10). After the addition of urea (0.65 g), the mixture was stirred continuously at room temperature for another 12 h. The resulting gel was transferred into an autoclave for further crystallization at 180 °C for four days. The as-synthesized products were collected by filtration, washed with water, dried in air and calcined at 550 °C for 4 h for removal of the organic template and transformation of the methyl groups into silanol groups. For comparison, s-S1-OH zeolite samples with different molar ratios of organosilane to the total amount of silica of 1/50 and 1/20 were synthesized via similar procedures using 2.74 g of TEOS and 0.04 g of DEMS for the 1/50 sample, and 2.66 g of TEOS and 0.1 g of DEMS for the 1/20 sample. In this work, unless specified otherwise, the s-S1-OH zeolite has a molar ratio of organosilane to the total amount of silica of 1/10.

Procedures for the synthesis of ferric carbide catalysts and other zeolites are provided in the Supplementary Information.

Catalytic tests. The LT-FTO reactions were performed using a stainless-steel fixed-bed flow reactor with an inner diameter of 10 mm. Typically, the Fe-based catalyst (0.5 g) was mixed with the zeolite (1.5 g) (where the catalyst granule size was 20–40 mesh unless stated otherwise), and was positioned in the middle of the reactor. Additional quartz sand was added to the reactor to fix the position of the catalyst bed. For each reaction, the catalyst was pre-reduced at 400 °C for 6 h in a pure H_2 flow (flow rate, 10 ml min^{-1}) at atmospheric pressure. After reduction, the reactor was cooled to 260 °C. Then, the feed gas (H_2/CO/Ar with a molar ratio of 45/45/10) with a flow rate of 20 ml min^{-1} was introduced into the reactor, and the reactor was pressurized gradually to 2 MPa. For comparison with different zeolites, the the amount of Na-FeC_x catalyst (0.5 g) was constant in the reactor, and the different zeolites with various amounts were changed. The GHSV (gas hourly space velocity) was calculated on the basis of the weight of the Fe-based catalyst, that is 2,400 ml per h per g FeC_x, and the zeolite promoter was not included.

The gaseous products from the reactor were analysed using two online gas chromatographs. One is a Fu Li-9790 gas chromatograph equipped with a TDX-01 column (3 m × 3 mm) and a thermal conductivity detector; the other is a chromatograph equipped with a capillary PLOT-Al_2O_3 column (50 m × 0.53 mm × 0.25 μm) and a flame ionization detector (FID). The liquid-phase products collected in the cold trap were analysed using an offline Shimadzu GC-14C gas chromatograph equipped with an HP-5 column (100 m × 0.32 mm × 0.25 μm) and an FID. Argon was used as an internal standard for calculating the CO conversion and product selectivities (CO_2 and CH_4) in the thermal-conductivity-detector analysis. Methane was used as an internal standard for calculating the product selectivities in the online FID analysis. The offline products were analysed using *n*-hexadecane as an internal standard. The conversions and product selectivities were calculated on the basis of the amount of total carbon atoms. At the beginning of each reaction, the ferric oxide would change into carbide, so the CO conversion and selectivity data were unstable during this activation period. For characterizing the performance of each catalyst, the catalytic data were recorded after reaction for more than 40 h to obtain stable CO conversion and product selectivity values, which was to ensure that the reaction had reached steady operation.

Data availability

The data that support the findings of this study are presented in the Letter and Supplementary Information, and are available from the corresponding authors upon reasonable request. Source data are provided with this paper.

Acknowledgements

We thank F. Chen for kind help in the transmission electron microscopy characterization. This work was supported by the National Key Research and Development Program of China (2021YFA1500404), the National Natural Science Foundation of China (U21B20101, 21932006, 22102143, 22032005 and 22125304) and the National Postdoctoral Program for Innovative Talents (BX20200291).

Author contributions

C.W. and W.F. performed the catalyst preparation, characterization and catalytic tests. Z. Liu and A.Z. performed the theoretical calculations and wrote the corresponding part. H.L., L.L., H.Z., X.Q., S.X. and Y.W. participated in the catalyst characterization. Z. Liao and Y.Y. provided helpful discussion and compiled the process package. X.C. performed the X-ray photoelectron spectroscopy characterization. L.W. and F.-S.X. designed the study, analysed the data and wrote the paper. All authors discussed the results and commented on the manuscript.

Competing interests

This work has been protected by a Chinese patent with the application number 202110715110.6. The authors C.W., W.F., L.W. and F.-S.X. were involved in this patent. The other authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41565-022-01154-9.

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Peer review information *Nature Nanotechnology* thanks Jingxiu Xie and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

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